

Exploratory Data Analysis (EDA) Report

1. Summary Statistics

Population (normalized)

- Min: 0.000
- Max: 1.000
- Mean: 0.162
- Median (50th percentile): 0.051
- Std Dev: 0.231
- 75% of countries have population values below 0.225

Area (km²) (normalized)

- Min: 0.000
- Max: 1.000
- Mean: 0.125
- Median (50th percentile): 0.028
- Std Dev: 0.206
- 75% of countries have area values below 0.135

Insight: Both variables are heavily right-skewed, with a small number of countries having very large population and area values compared to the rest.

2. Distribution Visualizations

Histograms

- Population: Most countries have small normalized population values. A few countries are outliers with high population.
- Area (km²): Similar to population, most countries have smaller areas, with a few significantly larger.

Scatter Plot: Population vs Area

- The scatter plot indicates a slight positive relationship: countries with higher populations may also have larger land areas, but it's not a strong linear trend.
- A cluster of countries has both small population and area, indicating a dense region in the lower-left quadrant.

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3. Boxplots

- Population: Shows many countries tightly grouped with several high-end outliers.
- Area: Similar pattern with notable outliers indicating extremely large countries.

Insight: Presence of high-end outliers in both population and area confirms the long tail in both distributions.

4. Correlation Analysis

Correlation Coefficient (Population vs Area): r approximately 0.35

Insight: There's a weak-to-moderate positive correlation between population and area. This suggests that while larger countries can have larger populations, it's not a strong or consistent pattern.

5. Key Takeaways

- The dataset is normalized and contains outliers, especially on the higher end of both population and area scales.
- Most countries fall in the lower ranges for both variables.
- Distributions are right-skewed.
- Weak positive correlation between population and area; geography is not a strong predictor of population size alone.
- Visualization helps clearly distinguish typical countries from the few outlier nations with massive area or population.